Short and sweet

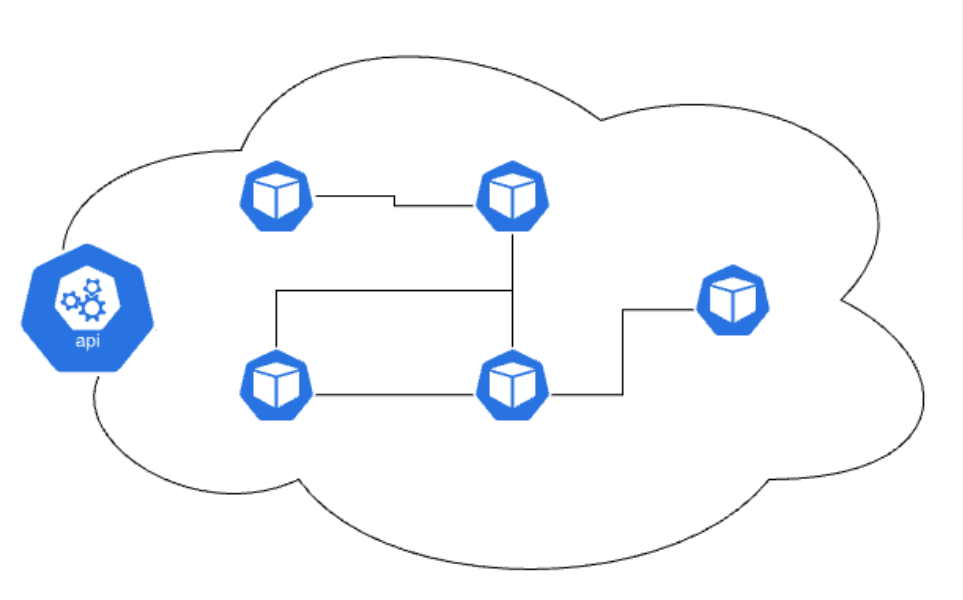
Kubernetes / Helm Basics

Kubernetes (K8s)

- "production grade container orchestrator"
- Features
  - High availability / Fault tolerance
  - Auto-scaling
  - Zero-downtime deployment
  - Load balancing

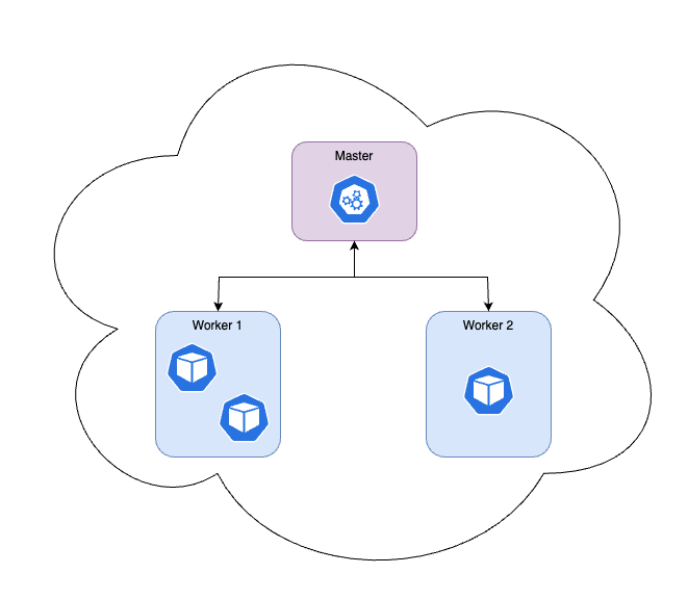
Kubernetes cluster

- The “Kubernetes application” in which containers can be deployed
- Consists of at least 1 node
- Concrete implementations:
  - AKS – Azure Kubernetes Service
  - EKS – Elastic Kubernetes Service (AWS)
  - GKE – Google Kubernetes Engine



Node

- A cluster of 1 ... n nodes
- Physical server or virtual machine
- Master node vs worker node
- Worker nodes run containers



Task - Nodes

How many nodes does the Dish dev cluster have?

```
kubectl get nodes
```

Namespace

- "folder" for all resources in a Kubernetes cluster
- only nodes are not assigned to a namespace

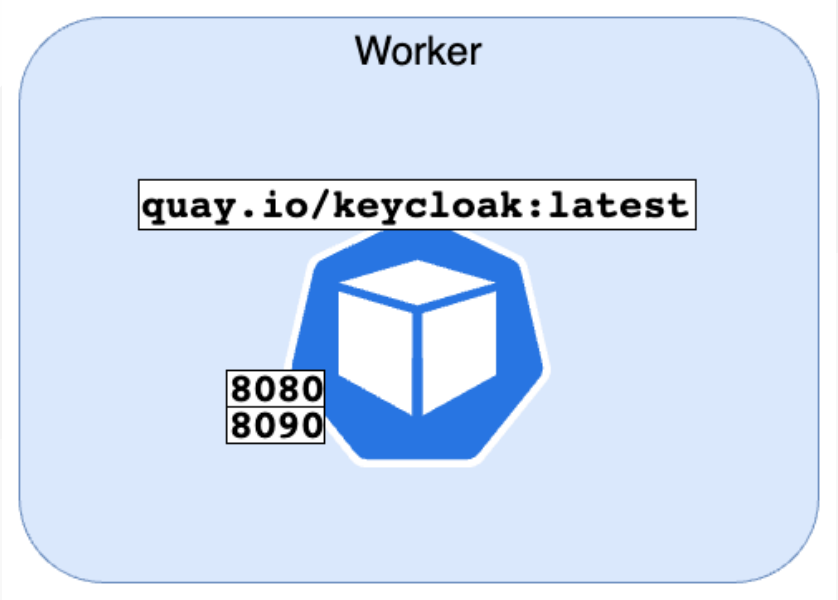
Task - Namespace

Create your own namespace in the Dev cluster:

```
kubectl create namespace <your-name>-test --dry-run=client -o yaml > namespace.yaml

kubectl apply -f namespace.yaml

kubectl get namespaces
```

Pod

- A deployed container
- Described by:
  - Container image
  - Environment variables
  - Ports
  - Protocols
  - …



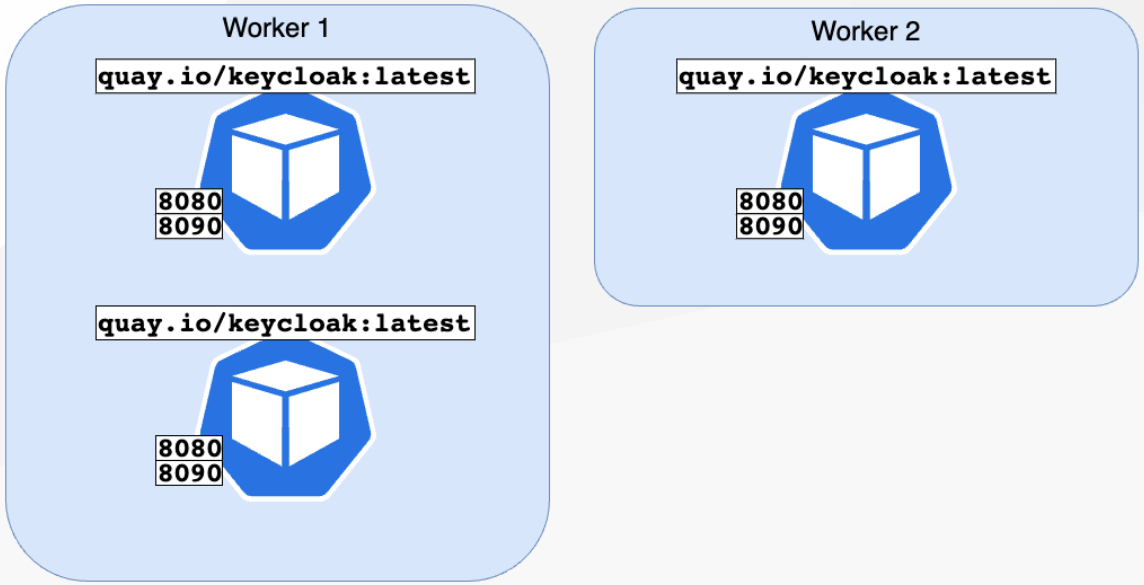
Task - Pod

Deploy a podinfo pod in your namespace:

```
kubectl run podinfo \
  --image=stefanprodan/podinfo \
  --restart=Never \
  --dry-run=client \
  --namespace=<your-namespace> \
  -o yaml > podinfo-pod.yaml

kubectl apply -f podinfo-pod.yaml -n <your-namespace>

kubectl get pods -n <your-namespace>

kubectl port-forward pod/podinfo 9898:9898 -n <your-namespace> # works also via OpenLens

# Open in browser http://localhost:9898

# Make the pod terminate and see how it restarts
curl localhost:9898/panic

kubectl get pods -n <your-namespace>

kubectl delete -f podinfo-pod.yaml
```

Deployment (deploy)

- Scaling an image
- update options



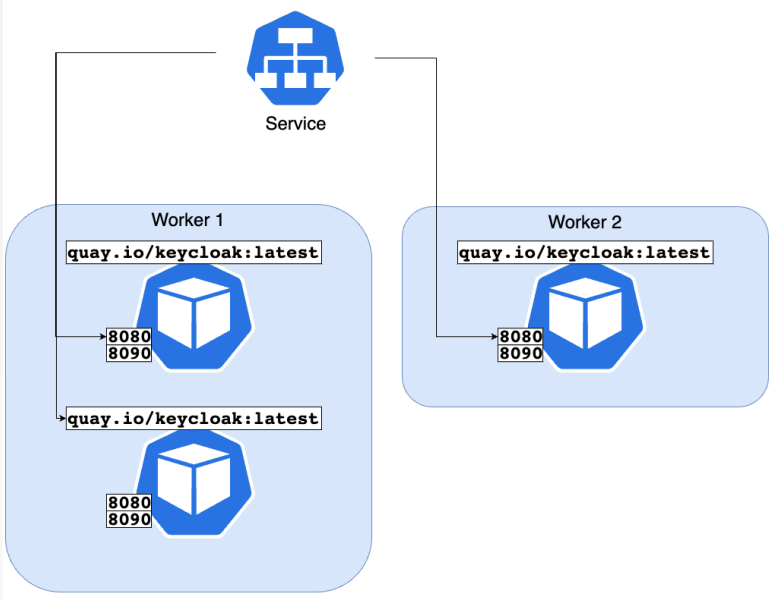
Task - Deployment

Deploy a podinfo pod as a deployment in your namespace:

```
kubectl create deployment podinfo-deploy \
  --image=stefanprodan/podinfo \
  --replicas=2 \
  --dry-run=client \
  --namespace=<your-namespace> \
  -o yaml > deployment.yaml

kubectl apply -f deployment.yaml -n <your-namespace>

kubectl get deployments -n <your-namespace>

kubectl get rs -n <your-namespace>

kubectl get pods -n <your-namespace>

kubectl scale deployment podinfo-deploy --replicas=3 -n <your-namespace>

# kubectl delete -f deployment.yaml
```

Service

- Provides a stable network endpoint to access Pods
- Pods are ephemeral (IP addresses change frequently)
- A Service solves this by:
  - Offering a fixed IP and DNS name
  - Load balancing traffic
  - Selecting Pods using labels
- Enables reliable communication between components inside a cluster



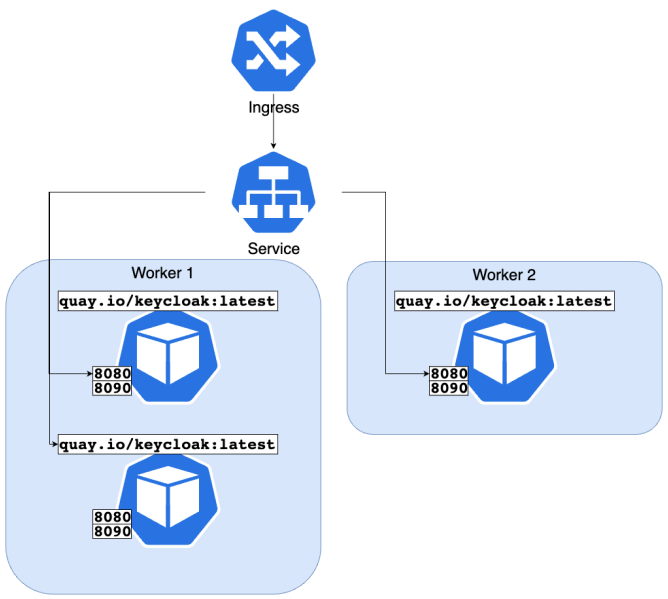
Task - Service

Deploy a K8s service for the existing deployment in your namespace:

```
kubectl expose deployment podinfo-deploy \
  --name=podinfo-svc \
  --port=9898 \
  --target-port=9898 \
  --type=ClusterIP \
  --namespace <your-namespace> \
  --dry-run=client \
  -o yaml > service.yaml

kubectl apply -f service.yaml -n <your-namespace>

kubectl get svc -n <your-namespace>

kubectl port-forward svc/podinfo-svc 9898:9898 -n <your-namespace>

# kubectl delete -f service.yaml
```

Ingress

- Manages external HTTP/HTTPS access to Services in a cluster
- Acts as a smart router for incoming traffic
- Provides:
  - URL-based routing
  - Host-based routing
  - Centralized entry point for applications



Task - Ingress

Deploy a K8s Ingress and point it to the service in your namespace:

```
# Update the file tasks/task-ingress/ingress.yaml

kubectl apply -f tasks/task-ingress/ingress.yaml -n <your-namespace>

kubectl get ingress -n <your-namespace>

kubectl port-forward svc/podinfo-svc 9898:9898 -n <your-namespace>

curl https://<ingress-host>/<your-namespace>/metrics

# kubectl delete -f ingress.yaml
```

Helm (Kubernetes Package Manager)

- Helm packages Kubernetes resources into reusable charts
- A Helm chart can deploy:
  - Deployment
  - Service
  - Ingress
- Makes applications:
  - Reproducible
  - Configurable
  - Easy to deploy across environments

Example folder structure

```
myapp/
├── Chart.yaml
├── values.yaml
└── templates/
    ├── deployment.yaml
    ├── service.yaml
    └── ingress.yaml
```

Task Helm

- inspect the files in `tasks/task-helm-chart`
- adapt to your needs

```
# Generate yaml files with Helm
helm template podinfo ./tasks/task-helm-chart --namespace <your-namespace> > output.yaml

# Or install right away
helm install podinfo ./tasks/task-helm-chart --namespace <your-namespace>
```

Task Cleanup

- cleanup everything you did:

```
kubectl delete namespace <your-namespace>
```

Kubernetes

- Container orchestration platform
- Automates deployment and scaling
- Provides self-healing and high availability
- Manages networking between services

Helm

- Package manager for Kubernetes
- Bundles resources into charts
- Enables reusable and configurable deployments
- Simplifies application installation and updates

Kubernetes – What we haven’t covered yet

- Jobs and CronJobs
- ConfigMaps and Secrets
- Network Policies and traffic control
- Stateful workloads and persistent storage

Helm – What we haven’t covered yet

- Chart dependencies
- Hooks for lifecycle automation
- Advanced templating features
- Environment-specific value management
- Release history and rollback capabilities

Opinion on Training

Your experience!